

1	Which of the following lung volumes cannot be measured by spirometry?
1	Tidal volume
2	Inspiratory reserve volume
3	Expiratory reserve volume
*	4 Residual volume

2	Which of the following muscles plays a key role during inspiration?
*	1 Diaphragm
	2 Scalene muscles
	3 Abdominal muscles
	4 Internal intercostal muscles

3	Under normal conditions, which of the following compartments has negative pressure?
	1 Atmosphere
	2 Air within the trachea
*	3 Pleural space
	4 Air within the pharynx

4	Which of the following is not part of the anatomical dead space?
*	1 Alveoli
	2 Trachea
	3 sinuses
	4 Bronchi

5	Which factor causes a rightward shift of the oxygen-hemoglobin dissociation curve?
	1 Decreased acidity
	2 Decreased temperature
*	3 Increased carbon dioxide concentration
	4 Decreased 2,3-bisphosphoglycerate (2,3-BPG)

6	What is the principal mechanism by which oxygen is transported in the blood?
	1 Binding to albumin
*	2 Binding to hemoglobin
	3 Dissolution in blood
	4 Diffusion

7	What is the principal mechanism by which carbon dioxide is transported in the blood?
	1 Binding to albumin
	2 Binding to hemoglobin
	3 Dissolution in blood
*	4 Transport as bicarbonate ions

دانشگاه علوم پزشکی:

آزمون:

دانشکده:

رشته تخصصی:

8	In spirometry, measurement of which volume requires <u>maximal effort</u> ?
	1 Tidal volume
*	2 Inspiratory reserve volume
	3 Functional residual capacity
	4 Residual volume
9	Which of the following is a <u>non-respiratory function</u> of the respiratory system?
*	1 Production of angiotensin II
	2 Transport of carbon dioxide from tissues to the lungs
	3 Regulation of respiratory gas concentrations
	4 Contraction of the diaphragm
10	In which region of the lung is physiological dead space present?
	1 Base of the lung
	2 Middle of the lung
*	3 Apex of the lung
	4 Throughout the lung
11	In which region of the lung is alveolar ventilation less than perfusion?
	1 Apex of the lung
*	2 Base of the lung
	3 They are always equal
	4 It never occurs
12	In respiratory tree, which of the following is part of the conducting zone of the respiratory system?
	1 Alveolar duct
	2 Alveolar sac
	3 Respiratory bronchiole
*	4 Bronchus
13	Which structure is inactive during quiet, normal inspiration?
*	1 Ventral respiratory group (VRG)
	2 Dorsal respiratory group (DRG)
	3 Diaphragm
	4 All of the above
14	Which center is responsible for the inspiratory "switch-off" signal?
*	1 Pneumotaxic center
	2 Dorsal respiratory group (DRG)
	3 Diaphragm
	4 Ventral respiratory group (VRG)

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15	What is the direct stimulus for the chemosensitive area?
	1 Oxygen
	2 Carbon dioxide
*	3 Hydrogen ion
	4 Bicarbonate ion

16	In environments with a high oxygen concentration, ...
*	1 Oxygen binds to hemoglobin
	2 Oxygen is released from hemoglobin
	3 No change occurs in the binding or release of oxygen from hemoglobin
	4 The rates of oxygen binding and release remain in equilibrium

17	Which factor increases lung compliance?
	1 Surface tension
*	2 Surfactant
	3 Lung collapse
	4 High lung volume

18	In which condition does a patient have obstructive lung disease?
*	1 Decreased FEV1/VC
	2 Increased FEV1/VC
	3 Unchanged FEV1/VC
	4 Cannot be assessed

19	The binding site of which molecule on hemoglobin is the same as that of oxygen?
	1 Carbon dioxide
*	2 Carbon monoxide
	3 Bicarbonate
	4 Hydrogen ion

20	In which of the following environments is the partial pressure of oxygen greatest?
*	1 Atmosphere
	2 Arterial blood
	3 Alveoli
	4 Venous blood

21	Which respiratory center is active during forced expiration?
	1 Pneumotaxic center
*	2 Ventral respiratory group (VRG)
	3 Dorsal respiratory group (DRG)
	4 Phrenic nerves

22	In which compartment is the partial pressure of carbon dioxide highest?	
*	1	Tissues
	2	Arterial blood
	3	Venous blood
	4	Alveoli

23	In microscopic structure Which of the following is observed bipolar Nuron Cell?	
	1	Vestibul of nasal cavity tissue
*	2	Olfactory zone tissue
	3	Respiratory zone tissue
	4	Tissue of Pharynx
Source:		
Description:		

24	In microscopic structure Which part of air duct respiratory system is observed bundle of elastic fiber?	
	1	Pharynx tissue
	2	Part of nasal cavity
*	3	Larynx tissue
	4	Part of bronchus

25	In microscopic structure which part of lobules Lung is observed simple Cuboidal epithelial tissue?	
	1	Alveolar
	2	bronchioles
*	3	Respiratory bronchiol
	4	Theracea

26	which part of microscopic structure Lung tissue is observed pneumocytes type III?	
*	1	In space alveolar
	2	Epithelial tissue of bronchus
	3	In space of trachea
	4	Epithelial tissue of Theracea

27	All of the following muscles are innervated by the recurrent laryngeal nerve except:	
	1	Lateral crico arytenoid
	2	Vocalis
*	3	Crico- thyroid
	4	Thyro-arytenoid

28	At the midclavicular line, the shadow of the lower side of the lung is located at which rib?	
	1	12th rib
	2	10th rib
	3	8th rib
*	4	6th rib

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29	All of the following are found in the middle meatus except.....?	
	1	Ethmoidal infundibulum
	2	Maxillary sinus opening
	3	Ethmoidal bulla
*	4	Nasolacrimal duct opening
30	Uncinate process is found in which of the following?	
	1	In the vomer bone
	2	In the medial wall of the nasal cavity
*	3	In the inferior surface of the ethmoidal labyrinth
	4	In the cribriform plate of the ethmoid
31	In a patient complaining of frequent and severe nosebleeds (epistaxis), examination reveals bleeding from the anterior-inferior region of the nasal septum. Which of the following arteries is not responsible for this bleeding?	
	1	Sphenopalatine
*	2	External nasal
	3	Superior Labial
	4	Greater palatine
32	Which of the following bones is involved in the roof and lateral wall of the nasal cavity?	
	1	Vomer
*	2	Ethmoid
	3	Lacrimal
	4	Palatine
33	Which of the following muscles attaches to the anterior surface of the pyramidal cartilage (arytenoid)?	
	1	Transvers Arytenoid
	2	Posterior Crico-arytenoid
*	3	Vocalis
	4	Oblique Arytenoid
34	The quadrangular membrane is attached to all of the following except...?	
*	1	Cricoid cartilage
	2	Thyroid cartilage
	3	Arythenoid cartilage
	4	Epiglottic cartilage
35	Which of the following nerves passes through the incisive canal?	
	1	External branch of the superior laryngeal nerve
*	2	Nasopalatine
	3	Internal branch of the superior laryngeal nerve
	4	Phrenic

36	Which of the following nerves passes through a hole in the thyrohyoid membrane?	
*	1	Internal branch of the superior laryngeal nerve
	2	Recurrent laryngeal
	3	Nasopalatine
	4	External branch of the superior laryngeal nerve
37	All of the following statement are correct except.....?	
	1	The azygos vein is adjacent to the medial surface of the right lung
*	2	The visceral pleura is innervated by the phrenic nerve
	3	The apex of the lung is located at the root of the neck
	4	The pulmonary vein is the most anterior element in the umbilicus of the left lung
38	All of the following elements are attached to the thyroid cartilage except the and the calyx	
	1	The true vocal fold
*	2	Aryepiglottic fold
	3	the false vocal cord
	4	inferior pharyngeal constrictor muscle